DO NOW · AO1 REVISION

# Precision, accuracy and uncertainty

**Working scientifically**
Both papers · Combined and Separate
9 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 1 ★ | **Define precision.** | How close repeated measurements are to each other. | ☐ |
| 2 ★ | **Define accuracy.** | How close a measurement is to the true value. | ☐ |
| 3 ★ | **Define the resolution of a measuring instrument.** | The smallest change the instrument can detect. | ☐ |
| 4 | **Define uncertainty.** | The interval within which the true value can be expected to lie. | ☐ |
| 5 | **State how the mean of a set of repeated readings is calculated.** | Add up the readings, then divide by the number of readings. | ☐ |
| 6 | **Define range.** | The maximum and minimum values of the independent or dependent variables. | ☐ |
| 7 ★ | **State how uncertainty is calculated from a set of repeated readings.** | Uncertainty = (largest reading − smallest reading) ÷ 2 | ☐ |
| 8 | **Three readings are close together but far from the true value. Describe them using the words precise and accurate.** | They are precise but not accurate. | ☐ |
| 9 | **Do repeated measurements help you judge precision or accuracy?** | Precision | ☐ |

© 2026 PanPhy Labs. All rights reserved.